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The Construction of Meaning

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Project A7: Focus and Extraction in Complex Constructions and Islands

# Subextraction from Present Participle Adjuncts in English

## Experimental Evidence

## 1 Theoretical background

Recent approaches to apparently transparent adjuncts (which are ruled out as ungrammatical by the *Condition on Extraction Domain*; Huang 1982) propose a variety of different factors that determine whether an adjunct is opaque or transparent. For example, Truswell (2007, 2011) locates this difference in the semantic component, arguing that the aspectual classes (Truswell 2007) and the agentivity of the two predicates (Truswell 2011) interact with *wh*-extraction. He proposes the *Single Event Grouping Condition* (Truswell 2011:157), which permits the extraction only when the two predicates can form a single event reading, which is conditioned by aspectual class membership and agentivity.

## 2 Experimental evidence

Two experiments were conducted to (i) show that aspectual class membership and extraction are two independent factors affecting acceptability (Experiment 1), and (ii) investigate whether there is a difference in acceptability with two types of matrix predicate that satisfy both of Truswell's conditions, but differ in whether the adjunct predicate can be seen as part of the matrix predicate's argument structure (Experiment 2). Both experiments test the *Interaction Hypothesis* in (1). The alternative is that effects of the choice of matrix predicate are independent of *wh*-extraction (2).

(1) **Interaction hypothesis:** Properties of the matrix predicate – such as aspectual class or argument structure – determine the strength of the effect of *wh*-extraction; e.g. Truswell (2007, 2011).

(2) **Independence hypothesis:** The choice of matrix predicate can have an independent effect on acceptability, but does not determine the strength of *wh*-extraction; e.g. Brown (2017).

### 2.1 Experiment 1

- This experiment tests the claim in Truswell (2007) that aspectual classes interact with *wh*-extraction, i.e. that some aspectual classes are affected by extraction more strongly than others.
- Items with telic and atelic matrix predicates in declaratives and with *wh*-extraction.
- 2×2 design, factors: MATRIX PREDICATE (*telic / atelic*) & STRUCTURE (*−wh / +wh*).
- 16 items, 39 participants, 7-point Likert scale.

(3) a. Dennis **came home** eating ice cream. [telic/−wh]
b. What did Dennis **come home** eating _? [telic/+wh]
c. Dennis *walked around* eating ice cream. [atelic/−wh]
d. What did Dennis *walk around* eating _? [atelic/+wh]

**Predictions:** the aspectual class has an effect only for one level of the factor STRUCTURE (Truswell 2007) → interaction between MATRIX PREDICATE & STRUCTURE.

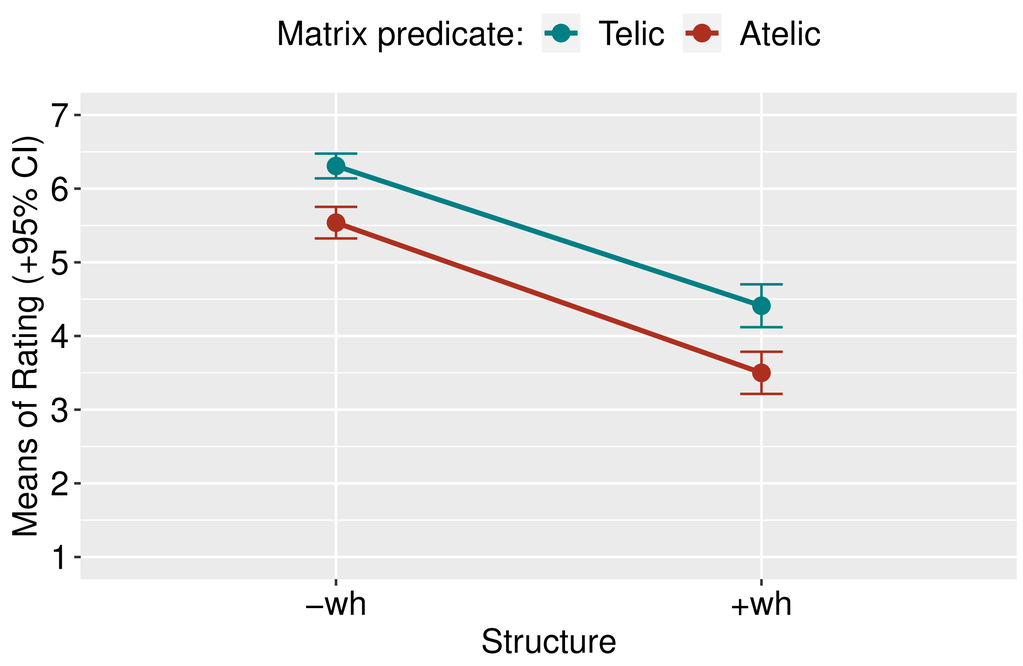



**Figure 1:** *Results from Experiment 1*

**Results**: Truswell's (2007) predictions are not confirmed: the aspectual class of the MATRIX PREDICATE has a significant effect on both levels of the factor STRUCTURE. The interaction between the two factors is not significant [$F_1(1, 38) = .577, p = .452, \eta^2 = .015$; $F_2(1, 15) = .115, p = .739, \eta^2 = .008$], meaning that they are two independent phenomena which affect acceptability in a linear-additive fashion. This speaks for an approach that separates licensing conditions on the gap inside the adjunct and (semantic) licensing conditions of the adjunct predicate (Brown 2017).

### 2.2 Experiment 2

- This experiment tests whether the potential argument structure status of the adjunct has an impact on the strength of *wh*-extraction.
- Items with *get*-predicates and achievements as matrix predicates.
- With *get*-predicates, the adjunct can fill an argument structure position of the matrix predicate (argument-adjunct; Grimshaw 1990), which is not possible with achievements (overlapping / depictive reading only).

(4) a. John got scared [watching a horror movie].
b. [Watching a horror movie] scared John.

- 2×2 design, factors: PREDICATE TYPE (*get / achievement*) & STRUCTURE (*−wh / +wh*).
- 16 items, 44 participants, 7-point Likert scale.

(5) a. Bill **got bored** listening to a politics lecture. [*get*-predicate/−wh]
b. What did Bill **get bored** listening to _? [*get*-predicate/+wh]
c. Bill *fell asleep* listening to a politics lecture. [Achievement/−wh]
d. What did Bill *fall asleep* listening to _? [Achievement/+wh]

**Prediction:** interaction between PREDICATE TYPE and STRUCTURE.




**Figure 2:** *Results from Experiment 2*

**Results:** The interaction between the two factors is not reliable [$F_1(1, 43) = 6.48, p < .05, \eta^2 = .131$; $F_2(1, 15) = 3.542, p = .0794, \eta^2 = .191$]; there is also no significant effect of VERB TYPE [$F_1(1, 43) = 2.201, p = .145, \eta^2 = .064$; $F_2(1, 15) = .334, p = .572, \eta^2 = .073$]. There is not enough evidence to claim that the potential argument structure status of the adjunct has an effect on the strength of the extraction. Note: Both predicate types (in this particular configuration with an atelic adjunct) are equally good under the *Single Event Grouping Condition* (Truswell 2011).

## 3 Discussion

- No evidence to support the *Interaction Hypothesis* (1).
- Suppport for the *Independence Hypothesis* (2) → independent licensing conditions for the adjunct (semantic) and for the gap inside the adjunct (syntax); Brown (2017).
- Next steps: focus on the licensing conditions for the adjunct predicate → how good are different types of matrix predicate with participial adjuncts?
- Can a more general notion of boundedness that goes beyond telicity and sub-stages explain the cases that Truswell (2007, 2011) struggles with? → (6).
- Can this explain why habitual readings of activities (7) seem to improve?

(6) a. What did John *appear / √arrive whistling _?
b. What did John walk *(around) whistling _?

(7) What does John always work whistling _?

## 4 Selected references

Brown, Jessica M. M. 2017. *Heads and adjuncts: An experimental study of subextraction from participials and coordination in English, German and Norwegian*: University of Cambridge dissertation.
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